

Disc classification for AI search

A deterministic, explainable formula specification for your multi-store catalog.

Version 0.1 | 28 September 2026

Recommended approach

Use the flight numbers to compute descriptive fields and a low-power accessibility score in the pipeline. Let the AI translate a shopper's request into those fields; do not let it invent flight behavior or recalculate the rules during search.

What changes from the earlier proposal

Earlier idea	Revised design
Turn + fade determines stability	Keep turn and fade separate. Add a two-axis shorthand that preserves a turn-and-fade profile instead of calling it straight.
One score maps to beginner / intermediate / advanced	Use a low-power access score plus a conservative beginner-use policy. A low score does not mean advanced-only.
Lighter is always better	Apply only a modest, capped driver-oriented weight adjustment. Do not penalize normal-weight putters and midranges.
Ease of distance is a second score	Drop the duplicate score. Flight ratings alone do not establish actual distance or universal controllability.

What the system will produce

Descriptive fields: power_band, turn_band, fade_band, glide_band and approx_stability. Ranking fields: access_model, access_variant and missing-weight bounds. Recommendation field: beginner_role. Operational fields: data_status, reason_codes and algorithm_version.

Scope and limits

The score targets lower-power throwing in ordinary, low-wind conditions, not putting performance, measured launch speed, tournament skill, or maximum distance. Compare candidates within the requested disc role rather than treating putters and distance drivers as substitutes.

Every coefficient, threshold and class boundary in this document is a proposed application heuristic, not an Innova formula or a statistically fitted model. A score of 80 is not an 80% probability of suitability. Manufacturer guidance supplies the direction of effects, not the numerical weights. [R1-R5]

Reading map: inputs (2); flight classes (3); core formulas (4-5); beginner policy (6); worked examples and checks (7-8); similarity and AI search (9); data contract (10); validation (11); sources (12).

1. Inputs and data quality

Compute the baseline once for each accepted flight-rating record, normally at manufacturer + mold level. Use a separate plastic/run-specific record when reliable published values differ. Join the baseline to individual store variants, then apply weight handling.

Input	Interpretation / rule
S, G, T, F	Speed, glide, turn, fade as finite numbers. Preserve decimals, including values such as -1.5.
w	An exact variant weight in grams. Keep a stated weight range as two endpoints, not an invented exact weight.
Manufacturer + model	Use the normalized identity. Preserve the source product and variant identifiers as strings.
Flight source and evidence	Keep the original tuple, provenance, attribution and any source conflicts. Do not combine four numbers from unrelated records.
Category and description	Useful supporting evidence. Missing type flags must not prevent calculation from valid flight numbers.

Supported numeric domain for v0.1

1 <= S <= 14	1 <= G <= 7
-5 <= T <= 1	0 <= F <= 5

These are the conventional ranges used by Innova and the supported domain of this prototype. Out-of-range values may be legitimate in another rating system: preserve them, flag them for review and withhold dependent scores. Do not clamp raw source data to force it into range. [R1]

For weights, use 80-220 g only as a broad pipeline review range, not a competition-legality rule. Invalid, non-finite or implausible weights become unavailable for adjustment; valid flight-based outputs remain available.

What the uploaded sample actually contains

Normalized model	Rows	Flight numbers	Weights
2025 Ace Race Prototype	14	All four missing	169-177 g
2024 Ace Race Prototype	28	All four missing	164-173 g
Anax	8	10 / 6 / 0 / 3	167-174 g

The file contains 50 variant rows, 5 product IDs and 3 normalized models. All 50 have weights; only 8 have complete flight numbers. This sample is not a coverage estimate for your full catalog. Counts above are calculated from the supplied JSON. [D1]

A concrete indexing issue

All eight Anax rows have all four category flags set to false despite valid flight numbers. Discraft calls the Anax an overstable fairway driver. Retain that sourced category when available; keep a separate speed-based power band. Do not silently turn every speed-10 disc into a distance-driver category. [D1, R6]

2. Describe flight on separate axes

Turn describes the early, faster portion of flight; fade describes the later finish. Innova cautions that actual flight depends on the throw and that ratings are not an exact prediction. Use the following classes as searchable approximations. [R1-R2]

Field	Exact v0.1 boundaries
power_band	low: $S \leq 5$; moderate: $5 < S \leq 9$; high: $9 < S \leq 12$; very_high: $S > 12$.
turn_band	high_turn: $T \leq -3$; moderate_turn: $-3 < T \leq -1$; mild_turn: $-1 < T < 0$; turn_resistant: $T \geq 0$.
fade_band	gentle: $F \leq 1$; moderate: $1 < F < 3$; strong: $F \geq 3$.
glide_band	low: $G \leq 3$; moderate: $3 < G < 5$; high: $G \geq 5$.

power_band is only a speed-based proxy, not a measured power requirement. A more negative turn can improve the access score without changing the published speed or this band.

Two-axis stability shorthand

Calculate approx_stability from valid T and F. These six cases partition the supported inputs without overlapping boundary values:

Condition	approx_stability
$F \geq 2$ and $T < -1$	turn_and_fade
$F \geq 4$ and $T \geq -1$	very_overstable
$2 \leq F < 4$ and $T \geq -1$	overstable
$F < 2$ and $T \leq -3$	very_understable
$F < 2$ and $-3 < T \leq -1.5$	understable
$F < 2$ and $T > -1.5$	neutral

Why there is no turn + fade score

$T = -3$ and $F = 3$ has the same sum as $T = 0$ and $F = 0$. The first suggests substantial early turn plus a substantial finish; the second does not. The rule keeps the first as turn_and_fade and the second as neutral, rather than collapsing both to a supposed straight flight.

The labels are app shorthand, not manufacturer-certified stability categories. "Neutral" does not guarantee a straight shot. Discraft's separate numeric Stability rating is a different field and must not be substituted for T or F. Its Anax page lists both systems. [R6]

When either required number is unavailable, approx_stability is null. Independently available bands may still be computed. Store description-derived claims in separate reported_* fields; do not fill numerical gaps from prose.

3. Core low-power access formula

This is one score, not separate beginner, power and distance scores with overlapping inputs. It is deliberately simple enough to implement in SQL or Python and to explain from component values. All constants below are tunable v0.1 design choices.

```
clip(x, lo, hi) = min(hi, max(lo, x))
c(x) = clip(x, 0, 1)
a = c((S - 5) / 2)
```

a is a driver-oriented blend: 0 at speeds 5 and below, 0.5 at speed 6, and 1 at speeds 7 and above. It allows neutral putters and mids to receive full turn-access credit without requiring negative turn.

Four component scores, each between 0 and 1

```
Speed access: A_s = c((13 - S) / 9)
Glide support: A_g = c((G - 2) / 4)
Gentle finish: A_f = c((3 - F) / 3)

Slow-disc turn support: U_s = c(1 - max(T, 0))
Driver turn support:    U_d = c((1 - T) / 3)
Combined turn support:  A_t = (1 - a)*U_s + a*U_d
```

Component	Contribution	Design intent
Speed access	50 points	Dominant factor. Full credit at $S \leq 4$; zero at $S \geq 13$.
Turn support	20 points	Neutral slow discs retain credit. Driver credit saturates at $T \leq -2$.
Gentle finish	15 points	Full credit at $F = 0$; zero at $F \geq 3$.
Glide support	15 points	No credit at $G \leq 2$; full credit at $G \geq 6$.

Unadjusted model score

```
A0 = 100 * (0.50*A_s + 0.20*A_t
            + 0.15*A_f + 0.15*A_g)
```

A0 is access_model. Do not round components or A0 during calculation. Round only the displayed value; examples use one decimal place, with halfway values rounded up. The formula requires all four valid flight numbers and no unresolved flight conflict.

Important behavior

More negative turn stops adding score after a point. It does not earn unlimited beginner credit. High-turn discs also pass through the separate use-policy checks on page 6, so easy turning is not confused with easy angle control.

Innova supports the broad direction of lower-speed accessibility, useful glide, and turn for lower-power players. The linear relationships, breakpoints, component weights and blend are our proposal, not measurements supplied by Innova. [R2]

4. Weight, ranges and missing values

Apply weight after the model score, at the store-variant level. Innova specifically notes that lighter drivers can help beginners, while heavier putters and midranges are commonly used from the start. This supports avoiding a blanket heavy-disc penalty. [R3]

Known exact weight

```
Weight adjustment: W = a * clip(0.5*(170 - w), -5, 10)
Variant score:      A = clip(A0 + W, 0, 100)
```

The 170 g reference and the 0.5-point-per-gram slope are proposed ranking settings, not physical constants. The adjustment is intentionally much smaller than the speed contribution. It never changes S, G, T, F or approx_stability.

Exact weight	Adjustment when a = 1	Effect at S <= 5
150 g or less	+10.0	0
155 g	+7.5	0
165 g	+2.5	0
170 g	0.0	0
175 g	-2.5	0
180 g or more	-5.0	0

At speed 6, multiply the middle column by 0.5. At speeds 5 and below, the algorithm makes no universal weight preference. That is a conservative modeling choice, not a claim that weight has no effect on those discs.

Unknown weight: preserve uncertainty instead of imputing

```
A_variant = null
A_low  = clip(A0 - 5*a, 0, 100)
A_high = clip(A0 + 10*a, 0, 100)
```

Keep access_model = A0 and has_exact_weight = false. A_low and A_high are bounds created by this formula's possible weight adjustment, not confidence intervals for real flight. For exact weights, set both bounds to A.

A retailer supplies a range [w_min, w_max]

```
A_variant = null
A_low  = score(w_max)
A_high = score(w_min)
score(w) = clip(A0 + a*clip(0.5*(170-w), -5, 10), 0, 100)
```

For a strict shopper weight range, require the full retailer interval to fit inside the requested interval. An overlapping interval is only a possible match and should be labeled separately. Missing weights must not satisfy an explicit weight requirement.

A missing glide, turn, fade or speed does not become zero. Do not reweight the remaining components to manufacture a complete score; keep A0 and all access scores null. Any independent descriptor that still has its required inputs can remain populated.

5. Beginner suitability is a policy layer

The score estimates lower-power accessibility. The policy below asks a narrower question: is this a reasonable default candidate for a beginner's ordinary throwing, a conditional option, or not a default? It does not assign the owner a skill level.

Apply these rules in order

```
If A0 is unavailable or flight data is conflicted:
    beginner_role = "unknown"

Else if A_low >= 70 AND S <= 9
    AND -2.5 <= T <= 0 AND F <= 2:
    beginner_role = "general_candidate"

Else if A_high >= 55 AND T <= 0 AND F <= 2:
    beginner_role = "conditional_candidate"

Else:
    beginner_role = "not_default"
```

The first rule uses the lower bound: unknown weight must not create a guaranteed general recommendation. The conditional rule uses the upper bound so that a potentially suitable light variant is not discarded; flag weight uncertainty when it matters.

Role	Meaning in the app
general_candidate	A conservative match for a generic beginner request within the appropriate disc category.
conditional_candidate	May fit a more specific need: lower-power distance, a stronger finish, or a disc requiring angle control. Explain the condition.
not_default	Do not make it a generic beginner recommendation. It may still be useful to any skill level for a particular shot.
unknown	The numeric algorithm cannot establish suitability. Standard search and clearly sourced descriptive claims remain available.

Do not convert the score into an exclusive skill label

Do not map 0-25 to "advanced," or interpret a high score as "not suitable for experts." Innova describes the Mamba as useful both for newer players and experienced players with slower arms. Skill, arm speed and intended shot are different variables. [R5]

If the UI needs a boolean, use true only for general_candidate; false for not_default; and null for conditional_candidate or unknown. Keep the richer role field as the source of truth.

Reason codes make the policy explainable

Examples: higher_speed when $S > 9$; high_turn when $T < -2.5$; stronger_finish when $F > 2$; and weight_not_exact when only a range or no weight is known. A near-threshold score can also be shown as borderline rather than as a firm promise.

6. Worked calculations

Your Anax record: 10 / 6 / 0 / 3, 167 g

These are actual sample inputs. The flight tuple matches Discraft's published Anax ratings. All four category flags in the sampled Anax rows are false, but that does not block scoring. [D1, R6]

```
a = 1
A_s = 3/9      A_t = 1/3
A_f = 0        A_g = 1
A0 = 16.6667 + 6.6667 + 0 + 15 = 38.3333
W = 0.5*(170 - 167) = 1.5
A = 39.8333 -> display 39.8
```

Outputs: power_band = high; turn_band = turn_resistant; fade_band = strong; approx_stability = overstable; beginner_role = not_default. This means "not our generic beginner pick," not "a beginner cannot use an Anax."

Mamba: 11 / 6 / -5 / 1, illustrative 155 g

The flight tuple is manufacturer-published; 155 g is a worked-example input, not a live inventory claim. Innova identifies lower-power uses for this disc. [R5]

```
a = 1
A_s = 2/9      A_t = 1
A_f = 2/3      A_g = 1
A0 = 11.1111 + 20 + 10 + 15 = 56.1111
W = 0.5*(170 - 155) = 7.5
A = 63.6111 -> display 63.6
```

Outputs: high speed-based power band, very_understable profile and conditional_candidate. The access score captures helpful turn and weight without pretending the disc has become a slow, neutral driver.

Mako3: 5 / 5 / 0 / 0, illustrative 180 g

The Star Mako3 tuple is 5 / 5 / 0 / 0. At speed 5, a = 0, so neutral turn receives full slow-disc credit and there is no generic heavy-weight penalty. [P2]

```
A0 = 44.4444 + 20 + 15 + 11.25 = 90.6944
W = 0
A = 90.6944 -> display 90.7
```

Result: neutral and general_candidate. This is a throwing-access assessment, not a putting recommendation or a prediction that it will outdistance a driver.

For the Anax with weight missing, A_variant would be null and the formula bounds would be 33.3-48.3. It remains not_default across the whole permitted weight adjustment.

7. Sanity checks against named discs

These are arithmetic and face-validity checks, not a measured accuracy study. Published flight tuples are cited in the source column. All weights are illustrative except the Anax's 167 g sample variant. Do not read the table as a ranking across disc categories.

Disc	S/G/T/F	g	Score	Beginner role	Source
DX Aviar	2/3/0/1	175	83.8	General	P1
Star Mako3	5/5/0/0	180	90.7	General	P2
DX Leopard	6/5/-2/1	165	81.4	General	P3
Star TeeBird	7/5/0/2	170	56.3	Conditional	P4
Star Roadrunner	9/5/-4/1	165	66.0	Conditional	P5
Mamba	11/6/-5/1	155	63.6	Conditional	R5
Star Wraith	11/5/-1/3	170	35.7	Not default	P6
Star Destroyer	12/5/-1/3	175	27.6	Not default	P7
Champion Firebird	9/3/0/4	175	30.1	Not default	P8
Anax (sample)	10/6/0/3	167	39.8	Not default	D1/R6

The Mamba and Roadrunner are not excluded just because of speed or extreme turn, but they are kept conditional. The Mako3 and Aviar are not penalized for ordinary heavier weights. The Firebird's low score does not erase its specialist uses. These are intended policy behaviors, not proof of recommendation quality.

Edge cases that must stay distinct

Input case	Required outcome
T = -3, F = 3 versus T = 0, F = 0	Different profiles: turn_and_fade versus neutral. No false straight-flight equivalence.
Valid speed and turn; fade missing	Keep power and turn bands; no stability class or complete access score.
Anax weight is 170-175 g	No exact score; access bounds are 35.8-38.3. Do not claim a 170 g exact match.
Slow disc with very negative turn	Can score highly for access, but fails the general-candidate turn gate.
Glide rises beyond 6 or turn falls below -2	No extra glide/driver-turn points after saturation. Preserve the raw values.
"0.0", "false" and null in the source	Parse deliberately. A real zero is valid; a string "false" is not truthy stock or a flight number.

Checks executed for this specification

A Python implementation reproduced the ten table scores and processed all 50 sample rows: 8 scored and 42 correctly remained unscored. It also passed 14,976 one-variable monotonic comparisons across randomly generated valid inputs, plus explicit null, range, saturation and profile-separation checks.

These checks establish formula consistency, not how well a person will throw any disc. The sample contains only one mold with complete ratings, so it cannot validate cross-brand recommendation performance.

8. Similar discs and AI search behavior

Similarity is a separate calculation

For a selected reference mold r , retrieve candidates with complete ratings and $|S - S_r| \leq 2$. Keep the requested category and all explicit shopper constraints. Rank the resulting molds by ascending distance:

$$D = 2 * \text{abs}(S - S_r) + \text{abs}(G - G_r) + 2 * \text{abs}(T - T_r) + 2 * \text{abs}(F - F_r)$$

Speed, turn and fade receive twice the weight of glide. These are provisional similarity weights, not a validated flight-distance metric. $D = 0$ means identical rating tuples, not identical flights. Do not convert D into a claimed match percentage.

Innova cautions against direct rating comparisons across brands and across speed classes. Exact-speed, same-brand comparisons are the stricter baseline; the wider speed window and cross-brand results are explicitly approximate suggestions. Label them accordingly and validate them with reviewed examples. [R1, R4]

Directional requests add constraints before ranking

Shopper request	Proposed interpretation
"Like this, easier to turn over"	$T \leq T_r - 1$ and $F \leq F_r$; then rank by D . Keep the reference speed window.
"Like this, less finish"	$F \leq F_r - 1$; then rank by D . Do not silently substitute a more negative turn.
"Beginner fairway driver"	Requested category plus general_candidate first; show conditional candidates as a clearly labeled expansion.
"Understable and under \$20"	approx_stability in {understable, very_understable}, then the strict price constraint. Mixed turn-and-fade is not an automatic match.
"A straight disc"	Offer low-turn/low-fade candidates, such as $-1 \leq T \leq 0$ and $F \leq 1$, in a suitable speed/category context; do not promise a straight flight.

How this fits your existing Typesense layer

Typesense Natural Language Search translates requests into q , filter_by and sort_by , with an optional custom system prompt. Publish the derived fields and document their meanings for that prompt. The model should use stored labels and scores, not invent missing numbers. [R7]

Reference-disc similarity requires an application-side lookup and calculation; an ordinary filter translation should not be expected to perform a cross-document join. Resolve the reference from your normalized catalog, retrieve candidates, and calculate D in the server or pipeline.

Protect the variant-level promise

Apply weight, price, stock and other hard constraints to the same variant before grouping results into product tiles. Display the lowest price among matching variants, not a cheaper sibling that fails the shopper's requirements. Keep parsed search parameters stable during pagination; do not needlessly reinterpret the same request on every page.

9. Pipeline and storage contract

Keep the deterministic transform between your accepted flight-number data and Typesense indexing. It does not need another LLM call for each product. AI remains useful for evidence extraction and query interpretation; rules own the numerical outputs.

Recommended output for the sample Anax variant

```
{
  "algorithm_version": "disc-classification-0.1",
  "data_status": "complete",
  "power_band": "high",
  "turn_band": "turn_resistant",
  "fade_band": "strong",
  "glide_band": "high",
  "approx_stability": "overstable",
  "access_model": 38.333333,
  "access_variant": 39.833333,
  "access_low": 39.833333,
  "access_high": 39.833333,
  "has_exact_weight": true,
  "beginner_role": "not_default",
  "reason_codes": ["higher_speed", "stronger_finish"]
}
```

Example numbers are shortened for readability; retain full computation precision. data_status = complete means the required numeric inputs are present and supported, not that real-world flight has been verified.

Preserve facts, claims and decisions separately

Layer	Store separately
Source facts	Accepted S/G/T/F tuple, normalized mold, plastic/run scope if known, exact weight or range, source URL/evidence and attribution.
Source claims	reported_stability, reported_skill and reported_use, each with evidence and source. Do not silently promote these to measured facts.
Rule outputs	The fields above, component values for diagnostics, input hash and algorithm version.
Human review	Approved overrides, scope, reason, reviewer and date. Overrides must not overwrite the original source tuple.

An upstream flight_confidence such as 0.98 measures whatever the existing extraction or matching pipeline defined; it is not 98% certainty about player suitability. Do not multiply the access score by that number. Preserve Try Discs attribution already carried in your sample. [D1]

Source IDs in this export are long integer-like strings. Keep them as strings end-to-end. Convert numeric and boolean fields explicitly; do not let generic string truthiness turn "false" into true. Missing score fields should remain absent/null with companion status fields, not be indexed as zero.

10. Validation before public recommendations

Use the formulas as a testable baseline

Version 0.1 is ready for controlled implementation and review, not for claiming validated personalization. Its strongest initial use is making existing numeric filters and clearly labeled candidate groups easier to request in natural language.

Validation area	Acceptance test
Arithmetic parity	The production SQL/Python/JavaScript implementation reproduces page 8 scores before display rounding. Test values exactly on every boundary.
Missingness	Absent values never become zero. One invalid field does not delete unrelated valid descriptors. Unknown-weight bounds enclose all permitted adjustments.
Monotonicity	Holding other inputs fixed: higher speed cannot increase access, higher fade cannot increase it, lower weight cannot reduce it, and supported extra glide/turn cannot reduce it.
Policy gates	High-turn or high-speed examples do not become unconditional beginner picks simply by being lightweight. Unknown flight numbers remain unknown.
Retrieval correctness	Every returned variant satisfies explicit filters. Grouped prices are computed only from matching variants. No empty type-flag set silently removes a valid mold.
Evidence and overrides	Conflicting sources are reviewable; copied retailer descriptions are not counted as independent confirmations. Overrides stay separate and versioned.

Build a small, independent judgment set

Have knowledgeable reviewers assess roughly 30-50 molds across manufacturers, speeds, flight profiles and selected plastics/weights. Specify the use case: generic beginner throwing, lower-power distance, or a particular shot. Collect conditional/unknown judgments rather than forcing every disc into one skill bucket.

Use part of that set to adjust the thresholds and reserve the rest for evaluation. Hold out whole molds, not random store variants of the same mold. Prioritize precision for general_candidate; show recall-expanding conditional results separately. Report results per brand and disc category before combining them.

Sensitivity matters more than a polished-looking score

Perturb the main coefficients and policy cutoffs around their proposed values, for example by about 10%, and inspect which labels change. Scores near 55 or 70 should be treated as borderline. Do not claim that a two-point difference predicts a meaningful performance advantage.

Keep these outside v0.1

Do not infer required throwing distance in feet, exact release velocity, forehand suitability, wind-speed tolerance or putting fit from the score. Do not add generic plastic offsets or pretend all runs of a mold fly identically. Player technique, release, wear and conditions matter, and this input set does not quantify them. [R4]

Practical starting point

Ship the descriptive bands, approximate stability, missing-data behavior and visible interpreted filters first. Run access scores and beginner-role labels in a review mode until the independent examples support them. The existing standard search remains available for unscored products.

Sources and provenance

Manufacturer guidance and official technical documentation were checked on 28 September 2026. Citations support the published definitions, cautions and input tuples; they do not endorse the proposed formulas. Titles below are clickable.

[R1] [Innova: Flight Ratings System](#)

Rating ranges, separate turn/fade, plastic effects, and comparison limits.

[R2] [Innova: Flight Numbers Made Simple \(21 April 2026\)](#)

Power, glide, turn and fade explained; flight depends on the throw.

[R3] [Innova: Disc Golf Guide - Why Weight Matters](#)

Lighter-driver guidance and the distinction for putters and midranges.

[R4] [Innova: Understanding Flight Numbers \(20 December 2022\)](#)

Same-speed/brand cautions; effects of technique, wear, wind and conditions.

[R5] [Innova: Disc Spotlight - The Mamba \(17 November 2025\)](#)

11/6/-5/1 and use cases spanning beginners, slower arms and experienced players.

[R6] [Discraft: Anax](#)

10/6/0/3, overstable fairway description and the distinct Discraft stability rating.

[R7] [Typesense v29: Natural Language Search](#)

Natural-language parameter translation and optional system prompt.

[P1] [Innova Factory Store: DX Aviar](#)

Published tuple: 2/3/0/1.

[P2] [Innova Factory Store: Star Mako3](#)

Published tuple: 5/5/0/0.

[P3] [Innova Factory Store: DX Leopard](#)

Published tuple: 6/5/-2/1.

[P4] [Innova Factory Store: Star TeeBird](#)

Published tuple: 7/5/0/2.

[P5] [Innova Factory Store: Star Roadrunner](#)

Published tuple: 9/5/-4/1.

[P6] [Innova Factory Store: Star Wraith](#)

Published tuple: 11/5/-1/3.

[P7] [Innova Factory Store: Star Destroyer](#)

Published tuple: 12/5/-1/3.

[P8] [Innova Factory Store: Champion Firebird](#)

Published tuple: 9/3/0/4.

Supplied data [D1]

`job_gLIvd9Ar8HKeuv5NQAXpAuEg-nqu.json`

The uploaded 50-row sample was inspected programmatically. The Anax inputs and attribution are in the final eight records; the first 42 records are Ace Race prototypes with null flight ratings. The file identifies its Anax flight source as `try_discs_exact`, with evidence dated 21 September 2026 and the attribution "Disc data by Try Discs (<https://trydiscs.com>)". No catalog-wide completeness or flight-performance conclusions are inferred from this sample.